Integrasi

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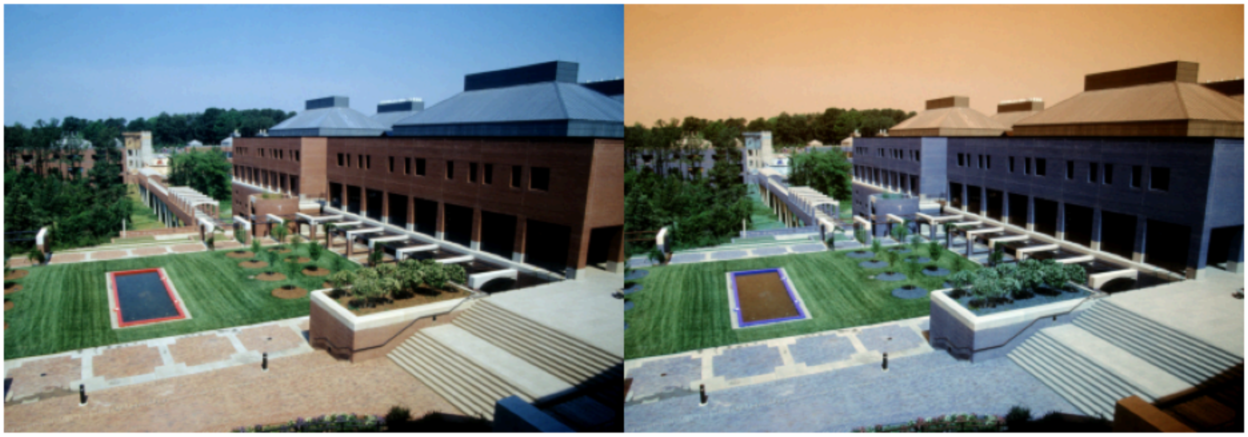
```
import numpy as np
import pandas as pd
import cv2 as cv
from skimage import io
from skimage import transform
from PIL import Image
import matplotlib.pyplot as plt
```

```
urls = ["https://iiif.lib.ncsu.edu/iiif/0052574/full/800,/0/default.jpg","https://iiif.lib.ncsu.edu/iiif/0016007/ful
```

```
for url in urls:

    image = io.imread(url)

    # Resize
    image = cv.resize(image, (0, 0), fx=0.5, fy=0.5)

    # Konversi warna
    image_2 = cv.cvtColor(image, cv.COLOR_BGR2RGB)

    # Gabungkan horizontal
    final_frame = cv.hconcat((image, image_2))

    # Display
    plt.figure(figsize=(12, 5))
    plt.imshow(final_frame)
    plt.axis("off")
    plt.show()

    print("\n")
```



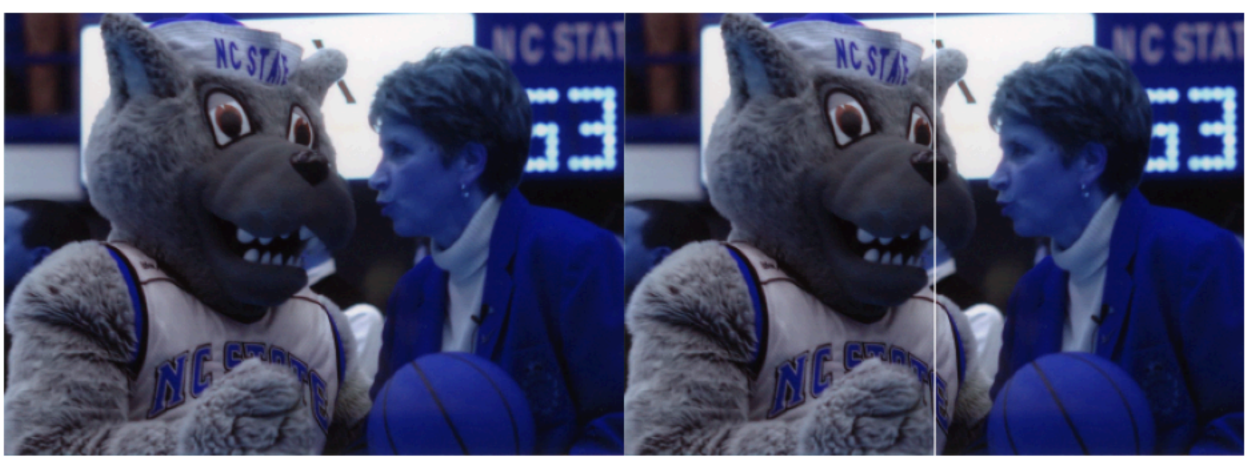
```
tinggi = image_2.shape[0]
lebar = image_2.shape[1]

print(f'Resolusi Panjang X Lebar: {tinggi} X {lebar}')

plt.figure(figsize=(12, 5))
plt.imshow(image_2)
plt.axis("off")
plt.show()
```

```
Resolusi Panjang X Lebar: 286 X 400
```

```
image_2 = cv.cvtColor(image, cv.COLOR_BGR2RGB)
image_3 = cv.cvtColor(image, cv.COLOR_BGR2RGB)

#membuat garis horizontal ditengah image
for y in range (lebar):
    image_3 [int((tinggi)/2),y] = [255,255,255]

final_frame = cv.hconcat((image_2, image_3))

plt.figure(figsize=(12, 5))
plt.imshow(final_frame)
plt.axis("off")
plt.show()
```

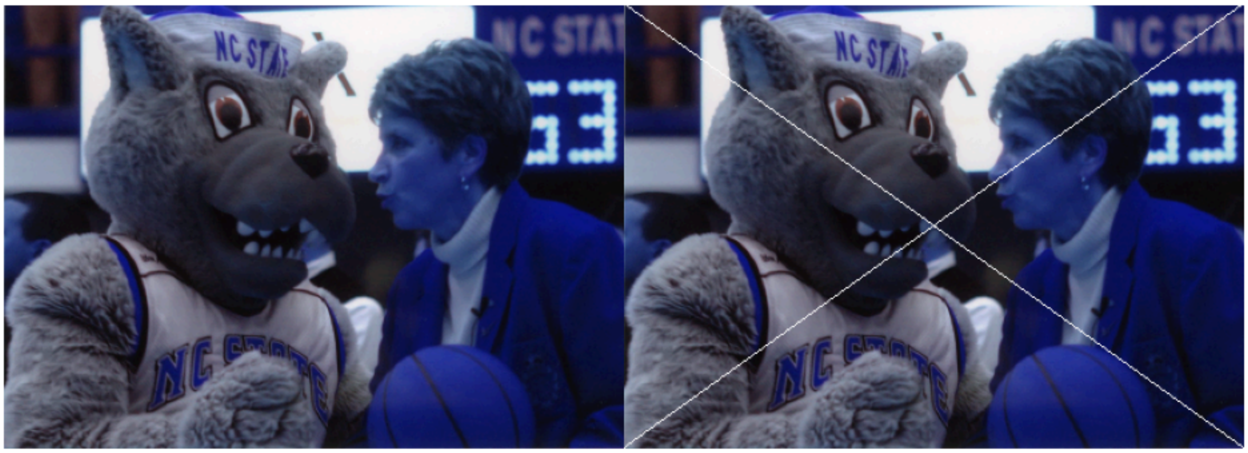
TUGAS PRAKTIKUM D2

```
image_2 = cv.cvtColor(image, cv.COLOR_BGR2RGB)
image_3 = cv.cvtColor(image, cv.COLOR_BGR2RGB)

#membuat garis Vertikal ditengah image
x = int(lebar / 2)

for y in range(tinggi):
    image_3[y, x] = [255, 255, 255]

final_frame = cv.hconcat((image_2, image_3))

plt.figure(figsize=(12, 5))
plt.imshow(final_frame)
plt.axis("off")
plt.show()
```

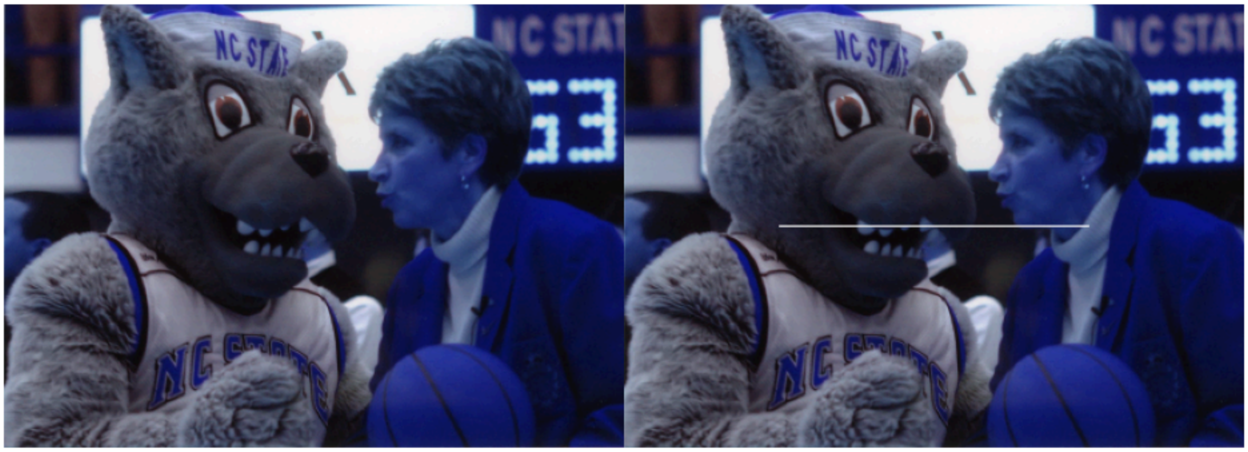


```
image_2 = cv.cvtColor(image, cv.COLOR_BGR2RGB)
image_3 = cv.cvtColor(image, cv.COLOR_BGR2RGB)
```

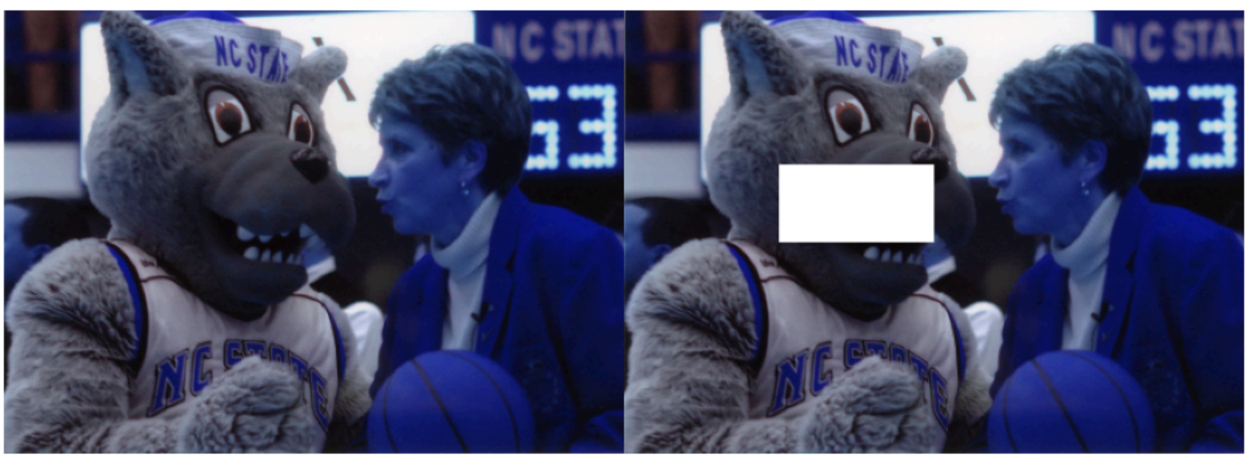
```
x = int(lebar / 2)

for x in range(lebar):
    y = int(tinggi - (x * tinggi / lebar) - 1)

    if 0 <= y < tinggi:
        image_3[y, x] = [255, 255, 255]

for x in range(lebar):
    y = int(x * tinggi / lebar)

    if 0 <= y < tinggi:
        image_3[y, x] = [255, 255, 255]

final_frame = cv.hconcat((image_2, image_3))

plt.figure(figsize=(12, 5))
plt.imshow(final_frame)
plt.axis("off")
plt.show()
```



```
image_2 = cv.cvtColor(image, cv.COLOR_BGR2RGB)
image_3 = cv.cvtColor(image, cv.COLOR_BGR2RGB)

tinggi, lebar, _ = image_3.shape

panjang_garis = 200
y = tinggi // 2

x_awal = (lebar - panjang_garis) // 2
x_akhir = x_awal + panjang_garis

image_3[y, x_awal:x_akhir] = [255, 255, 255]

final_frame = cv.hconcat((image_2, image_3))

plt.figure(figsize=(12, 5))
plt.imshow(final_frame)
plt.axis("off")
plt.show()
```



```
image_2 = cv.cvtColor(image, cv.COLOR_BGR2RGB)
image_3 = cv.cvtColor(image, cv.COLOR_BGR2RGB)

x_awal = 100
y_awal = 100

x_akhir = 200
y_akhir = 150

for y in range(y_awal, y_akhir):
    for x in range(x_awal, x_akhir):
        image_3[y, x] = [255, 255, 255]

final_frame = cv.hconcat((image_2, image_3))

plt.figure(figsize=(12, 5))
plt.imshow(final_frame)
plt.axis("off")
plt.show()
```



Pengolahan Citra dan Visi Komputer – Jurusan Teknologi Informasi

Pertanyaan

1. Google Colab

Jelaskan, mengapa pada modul praktikum ini eksekusi kode Python dilakukan menggunakan **Google Colab**?

2. Library pada Praktikum

Jelaskan mengenai kegunaan setiap **library** pada praktikum langkah ke-8.

Apakah semua library tersebut harus digunakan dalam praktikum sesi ini? Jelaskan!

3. Uji Coba Langkah ke-9

Pada uji coba langkah ke-9 terdapat potongan kode program sebagai berikut:

```
    image = cv.resize(image, (0, 0), fx=0.5, fy=0.5)
```

Apa kegunaan kode program tersebut? Dan apa pengaruhnya jika kode tersebut tidak dilakukan?

4. Nilai `[255, 255, 255]`

Perhatikan potongan kode program berikut:

```
for y in range (lebar):
    image_3 [int((tinggi)/2),y] = [255,255,255]
```

Apakah kegunaan kode:

```
[255, 255, 255]
```

Jelaskan!

5. Pixel dan Resolusi Gambar

Jelaskan keterkaitan antara **pixel** dan **resolusi gambar**, baik pada gambar dengan resolusi tinggi maupun rendah.

Jawaban

1.Google Colab

Karena Google Colab menyediakan GPU dan juga environment yang terinstall library yang dibutuhkan untuk praktikum ini.

2.Library pada Praktikum